



Artifact Repository Manager with Nexus

Key Takeaways

Introduction to Artifact Repository Manager

What is an Artifact Repository?

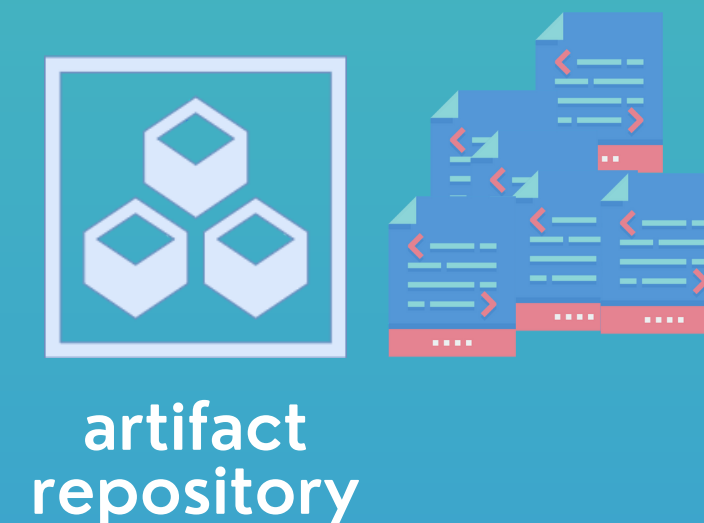
- **Storage of build artifacts** produced by continuous integration which makes them available for automated deployment to different deployment environments
- Provides a **central location**

- Artifacts are applications built into a **single file**
- There are different artifact **formats**:

JAR, WAR, ZIP, TAR, etc.



- Artifact repository needs to **support this specific format**
- Repository for each file/artifact type



What is an Artifact Repository Manager?

- Can **store many different artifact types**
- Which is great, because as a company you often produce different types of artifacts



Instead of **having different repositories for each** artifact type

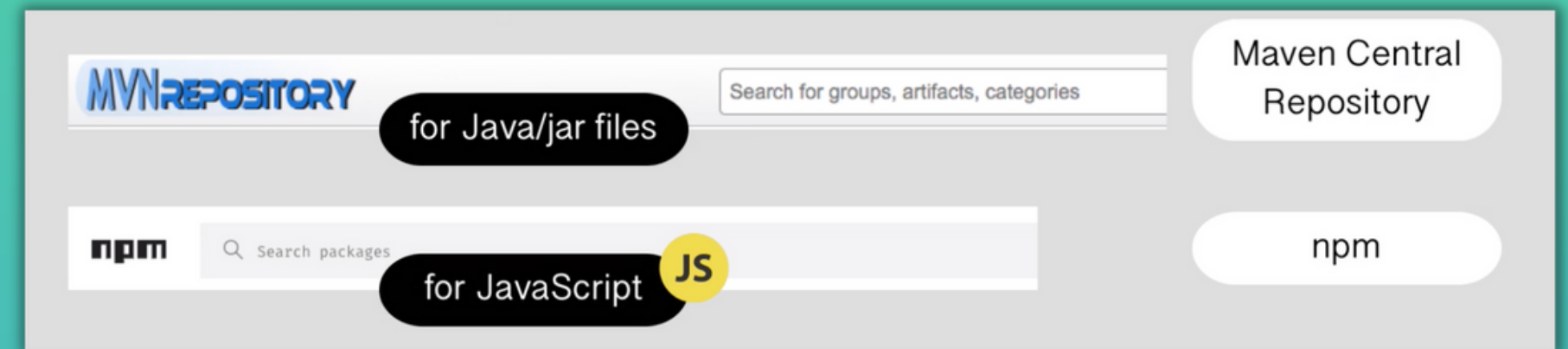


Just 1 repository for managing all your different artifact types

Public vs Private Artifact Repository Managers

Public Repository Managers

- There are public repository managers
- For example libraries/frameworks you use as a dependency
- You can make own project publically available there:



Public

Nexus - one of the most popular repository managers

- It's a private central store
- You can **upload** and **store** different build artifacts
- **Retrieve** (download) artifacts later
- For internal usage in your company



Private

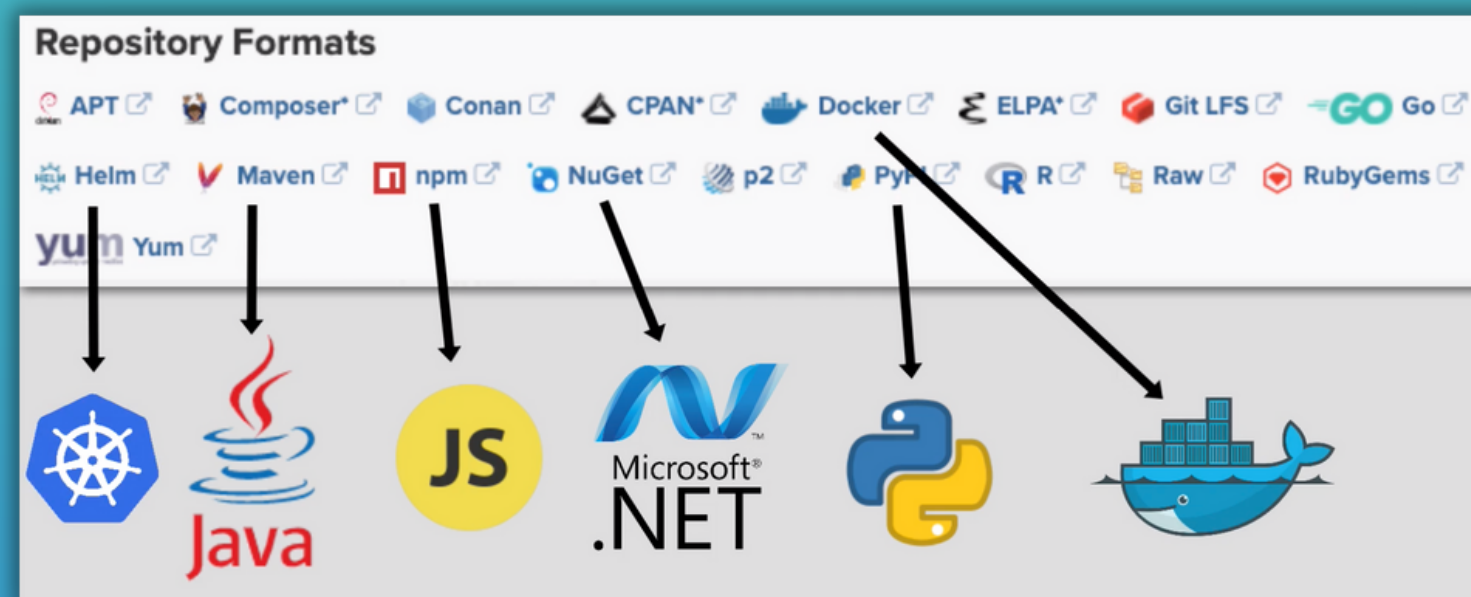
Nexus - Artifact Repository Manager

- Has open source and commercial version
- Host own repositories
- Proxy repositories (intermediary to another repository):



Fetch everything from 1 place (Nexus):
Company internal and public artifacts

- You can have multiple repositories for different formats or different teams in your company
- **Supported formats** in Nexus:



Features of Repository Manager



Integration with LDAP



Flexible and powerful REST API for integration with other tools

Important, because you need to integrate it into your CI/CD Pipeline



Backup and Restore



Multi-Format Support (different file types - zip, tar, docker etc)



Metadata Tagging (labelling and tagging artifacts)



Cleanup Policies (automatically delete files that match condition)



Search functionality (across projects, artifact repos etc.)



User token support for system user authentication

Install and Run Nexus on a Cloud Server

Install and Run Nexus on a Cloud Server - 1

Summary of Steps - 1st Part:

1. **Create Ubuntu Server (Droplet)** - min 4GB RAM, 2 CPUs, 160 GB SSD Disk
2. **Open SSH port 22**
3. **Install Java 8**
4. **Download and Install Nexus**
5. **Create "nexus" user and group** - Best Practice: Run applications with own user

Install and Run Nexus on a Cloud Server - 2

With Nexus installation you get:

```
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt# ls
digitalocean nexus-3.53.0-01 nexus-3.53.0-01-unix.tar.gz sonatype-work
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt# ls nexus-3.53.0-01/
NOTICE.txt OSS-LICENSE.txt PRO-LICENSE.txt bin deploy etc lib public
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt# ls sonatype-work/nexus3/
blobs clean_cache log tmp
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt#
```

Sonatype-work:

contains own config for Nexus and data

- Subdirectories depending on your Nexus configuration
- IP address that accessed Nexus
- Logs of Nexus App
- Your uploaded files and metadata
- You can use this folder for backup



Nexus folder:

contains runtime and application of Nexus

Install and Run Nexus on a Cloud Server - 3

Summary of Steps - 2nd Part:

1. Make Nexus owner of the folders
2. Start with nexus user
3. Open port 8081 with firewall
4. Access from browser

Starting Nexus...

- Services should not run with root user permissions
- So best practice: create own user for service (like in this case **nexus**)
- Should only have the permission for that specific service



Install and Run Nexus on a Cloud Server - 4

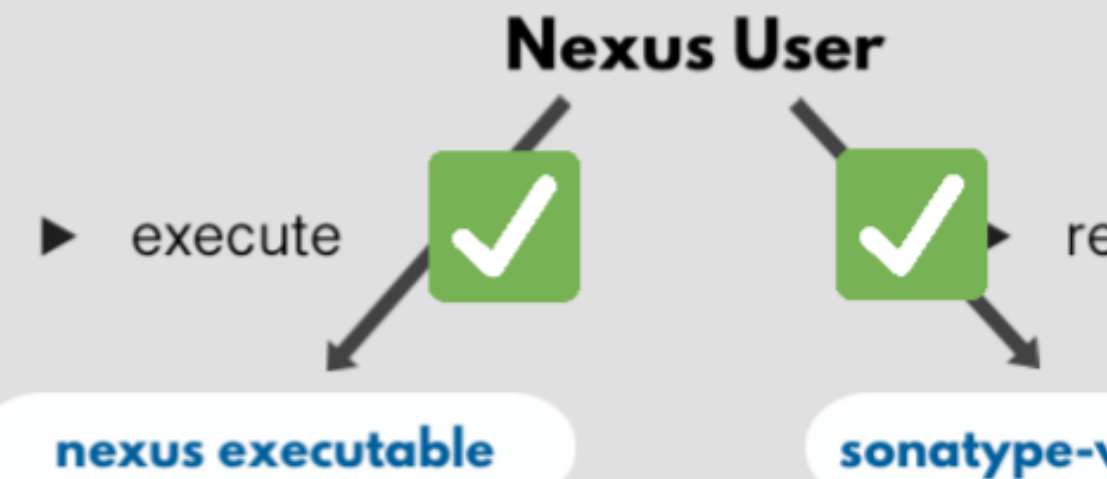
Run Nexus...

1. Make Nexus own the folders

2. Start with nexus user

```
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt# ls -l
total 205600
drwxr-xr-x  4 root root    4096 May  3 16:07 digitalocean
drwxr-xr-x 10 root root    4096 May  3 16:16 nexus-3.53.0-01
-rw-r--r--  1 root root 210520190 May  2 21:37 nexus-3.53.0-01-unix.tar.gz
drwxr-xr-x  3 root root    4096 May  3 16:16 sonatype-work
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt# chown -R nexus:nexus nexus-3.53.0-01
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt# chown -R nexus:nexus sonatype-work
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt# ls -l
total 205600
drwxr-xr-x  4 root root    4096 May  3 16:07 digitalocean
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-rw-r--r--  1 root root 210520190 May  2 21:37 nexus-3.53.0-01-unix.tar.gz
drwxr-xr-x  3 nexus nexus  4096 May  3 16:16 sonatype-work
root@ubuntu-s-4vcpu-8gb-fra1-01:/opt#
```

```
nexus@ubuntu-s-4vcpu-8gb-fra1-01:~$ /opt/nexus-3.53.0-01/bin/nexus start
Starting nexus
nexus@ubuntu-s-4vcpu-8gb-fra1-01:~$ ps aux | grep nexus
root      2543   0.0  0.0  9744 3984 pts/0    S    16:23   0:00 su - nexus
nexus     2544   0.1  0.0   9084 5352 pts/0    S    16:23   0:00 -bash
nexus     2760   2.73  9.0 6346460 736156 pts/0  Sl   16:23   0:32 /usr/lib/jvm/java-8-openjdk-amd64/jre/bin/java -server -Dinstall4j.jvmDir=/usr/lib/jvm/java-8-openjdk-amd64/jre -Dexe4j.moduleName=/opt/nexus-3.53.0-01/bin/nexus -XX:+UnlockDiagnosticVMOptions -Dinstall4j.launcherId=245 -Dinstall4j.swt=false -Di4jv=0 -Di4jv=0 -Di4jv=0 -Di4jv=0 -Di4jv=0 -Xmx2703m -Xms2703m -XX:MaxDirectMemorySize=2703m -XX:+UnlockDiagnosticVMOptions -XX:+LogVMOutput -XX:LogFile=../sonatype-work/nexus3/log/jvm.log -XX:-OmitStackTraceInFastThrow -Djava.net.preferIPv4Stack=true -Dkaraf.home=. -Dkaraf.base=. -Dkaraf.etc=etc/karaf -Djava.util.logging.config.file=etc/karaf/java.util.logging.properties -Dkaraf.data=../sonatype-work/nexus3 -Dkaraf.log=../sonatype-work/nexus3/log -Djava.io.tmpdir=../sonatype-work/nexus3/tmp -Dkaraf.startLocalConsole=false -Djdk.tls.ephemeralDHKeySize=2048 -Djava.endorsed.dirs=lib/endorsed -Di4j.vpt=true -classpath /opt/nexus-3.53.0-01/.install4j/i4jruntime.jar:/opt/nexus-3.53.0-01/lib/boot/nexus-main.jar:/opt/nexus-3.53.0-01/lib/boot/activation-1.1.1.jar:/opt/nexus-3.53.0-01/lib/boot/jakarta.xml.bind-api-2.3.3.jar:/opt/nexus-3.53.0-01/lib/boot/jaxb-runtime-2.3.3.jar:/opt/nexus-3.53.0-01/lib/boot/txw2-2.3.3.jar:/opt/nexus-3.53.0-01/lib/boot/istack-commons-runtime-3.0.10.jar:/opt/nexus-3.53.0-01/lib/boot/org.apache.karaf.main-4.3.9.jar:/opt/nexus-3.53.0-01/lib/boot/osgi.core-7.0.0.jar:/opt/nexus-3.53.0-01/lib/boot/org.apache.karaf.specs.activator-4.3.9.jar:/opt/nexus-3.53.0-01/lib/boot/org.apache.karaf.diagnostic.boot-4.3.9.jar:/opt/nexus-3.53.0-01/lib/boot/org.apache.karaf.jaas.boot-4.3.9.jar com.install4j.runtime.launcher.UnixLauncher start 9d17dc87 0 0 org.sonatype.nexus.karaf.NexusMain
nexus     3019   0.0  0.0  10264 3192 pts/0    R+   16:23   0:00 ps aux
nexus     3020   0.0  0.0   6852 2096 pts/0    S+   16:23   0:00 grep --color=auto nexus
nexus@ubuntu-s-4vcpu-8gb-fra1-01:~$
```



Install and Run Nexus on a Cloud Server - 5

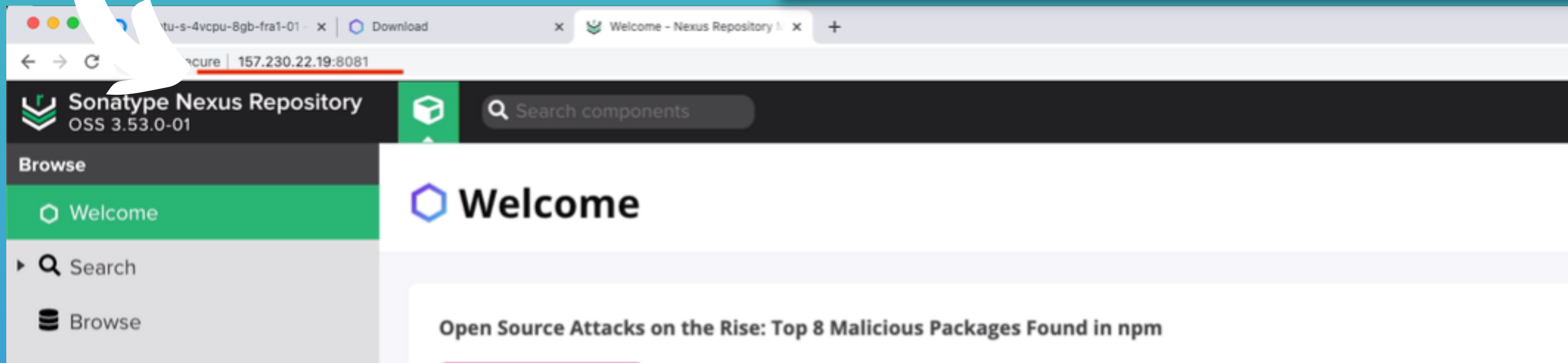
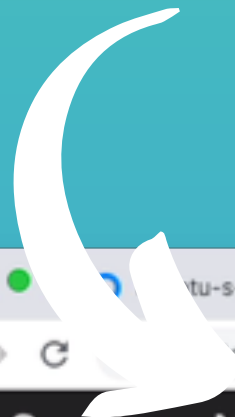
Run Nexus...

3. Open port 8081 on Firewall



```
nexus@ubuntu-s-4vcpu-8gb-fra1-01:~$ ps aux |
root      2543   0.0   0.0   9744   3984 pts/0
nexus     2544   0.0   0.0   9084   5352 pts/0
nexus     2760   24.0 21.6 6398276 1758736 p
er -Dinstall4j.jvmDir=/usr/lib/jvm/java-8-op
ckDiagnosticVMOptions -Dinstall4j.launcherId
s2703m -Xmx2703m -XX:MaxDirectMemorySize=270
work/nexus3/log/jvm.log -XX:-OmitStackTraceI
karaf.etc=etc/karaf -Djava.util.logging.conf
k/nexus3 -Dkaraf.log=../sonatype-work/nexus3
=false -Djdk.tls.ephemeralDHKeySize=2048 -Dj
01/.install4j/i4jruntime.jar:/opt/nexus-3.53
1.jar:/opt/nexus-3.53.0-01/lib/boot/jakarta.
:/opt/nexus-3.53.0-01/lib/boot/txw2-2.3.3.ja
s-3.53.0-01/lib/boot/org.apache.karaf.main-4
0-01/lib/boot/org.apache.karaf.specs.activat
-4.3.9.jar:/opt/nexus-3.53.0-01/lib/boot/org
er start 9d17dc87 0 0 org.sonatype.nexus.kar
nexus     3356   0.0   0.0  10264   3256 pts/0
nexus     3357   0.0   0.0   6852   2064 pts/0
nexus@ubuntu-s-4vcpu-8gb-fra1-01:~$ netstat
(Not all processes could be identified, non-
will not be shown, you would have to be roo
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address
tcp      0      0 127.0.0.1:45147
tcp      0      0 127.0.0.53:53
tcp      0      0 0.0.0.0:8081
tcp      0      0 127.0.0.54:53
```

4. Access from browser



Access Nexus from Browser



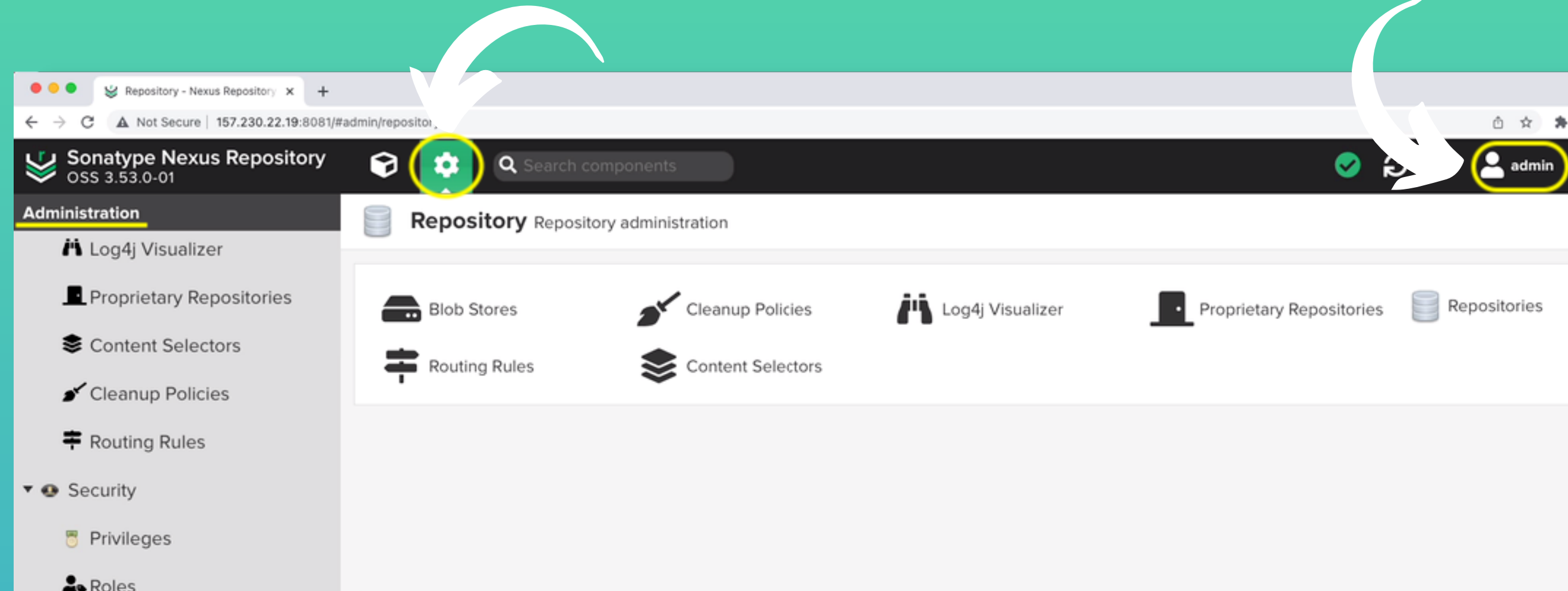
Open port 8081 !

Overview of Nexus

Login & Default Repositories

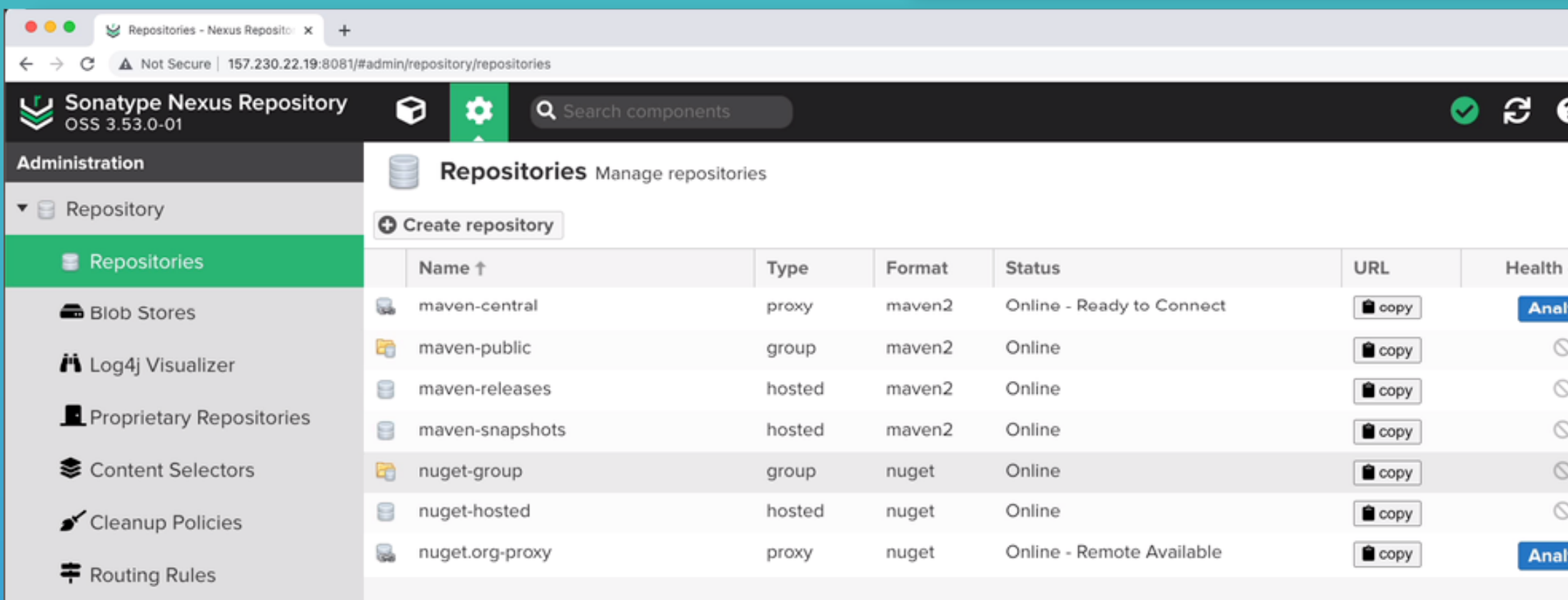
Login

- With **default "admin" user** that automatically gets created to administer Nexus



Default Repositories

- You get default repositories with different **types** when you install Nexus



Repository Types & Formats

Repository Types

- **Proxy:** intermediary to another repository
- **Group:** containing a group of repositories
- **Hosted:** for hosting company-internal artifacts



Each technology (python, java, docker) has its artifact type (jar, image), which can be saved in its own artifact repository format

Name ↑	Type	Format
maven-central	proxy	maven2
maven-public	group	maven2
maven-releases	hosted	maven2
maven-snapshots	hosted	maven2
nuget-group	group	nuget
nuget-hosted	hosted	nuget
nuget.org-proxy	proxy	nuget

Repository Formats

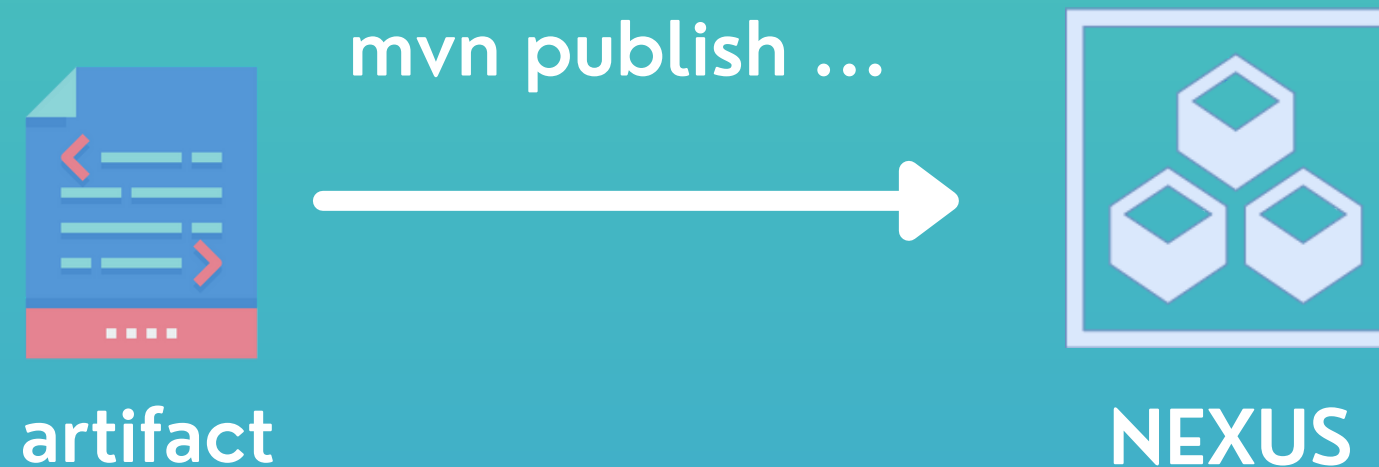
- **maven2:** for java jar artifacts
- **nuget:** for .NET packages
- **docker:** for Docker images

Publish artifact to repository

You can push or "PUBLISH" artifacts to Nexus with different tools

- Using built-in commands of build tools, like Maven or Gradle
- For that, you need to **configure** each build tool **with Nexus credentials and address**

X X X X X



Nexus REST API

You can also **interact with Nexus using its REST API** to do multiple things:

- query for different information:
- download an artifact
- upload an artifact

list of repositories

list of components

...

How to interact?

- Using a tool like curl or wget to execute http request
- You need to provide username and credentials of a Nexus user

```
nana@macbook /Users/nana  
% curl -u user:pwd -X GET 'http://157.230.22.19:8081/service/rest/v1/repositories'
```

```
nana@macbook /Users/nana  
% curl -u nana:techworld -X GET 'http://157.230.22.19:8081/service/rest/v1/components?repository=maven-snapshots'
```

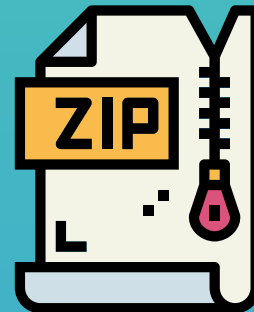


To use Nexus API or any API:
You should **always reference the documentation**. Not learn by heart,
because 1) API's change 2) there are a lot of endpoints

Component vs Asset

Components

- **abstract**
- what we are uploading
- term "component" refers to any type or format



Assets

- **actual physical packages/files**
- 1 component = 1 or more assets

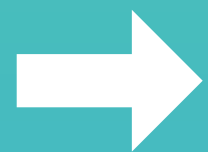
Cleanup Policies & Scheduled Tasks

Cleanup Policies

- Clean up old artifacts to make space. You can define logic like:



Delete all artifacts older than 30 days



Delete all artifacts, which haven't been used/downloaded for more than 10 days

- You can create individual cleanup policy for each repository



Scheduled Tasks

- Nexus allows you to schedule tasks that will be applied to all or a specific repository on a configurable schedule